

Hello everyone, and welcome to BLA 2 of my AWS Machine Learning Engineer Associate MLA-C02 learning series. BLA 1 introduced the basic ideas behind AWS Cloud, machine learning, and the complete AI and machine learning lifecycle. In BLA 2, I am moving into the first official exam domain, which is Data Preparation for Machine Learning and AI. This domain is important because every model depends on the data that reaches it.

This first presentation focuses on collecting and storing data. I will explain where machine learning data can come from, the difference between batch and streaming ingestion, how AWS storage services support different access patterns, and why file formats matter. I will also discuss structured, unstructured, and multimodal data, followed by the checkpoints that make an ingestion pipeline dependable. This presentation is conceptual. My goal is to understand the decisions an ML engineer must make before data transformation, and model development can begin.

The MLA-C02 exam gives Data Preparation for ML and AI twenty-eight percent of the scored content. It is the largest of the four exam domains. The first task in this domain is collecting and storing data. The engineer must know who owns the data, whether it can legally and ethically be used, how often it changes, and how quickly downstream systems need it.

Relevant data must represent the problem that the model will solve. Permitted data means the organization has the authority to collect, store, and process it. Accessible data means that authorized services and users can retrieve it when needed.

Two major ingestion patterns are batch ingestion and streaming ingestion. Batch ingestion moves data on a schedule or in groups. For example, a company may export sales records every night, upload a monthly customer file, or collect a large historical dataset before training a model. Batch processing works well when a delay is acceptable and when the workload benefits from processing many records together.

Streaming ingestion handles events as they arrive. Examples include website activity, sensor readings, application logs, stock transactions, and changes in customer behavior. On AWS, services such as Amazon Kinesis Data Streams and Amazon Managed Streaming for Apache Kafka can carry continuous event data. Apache Flink can process streaming records and calculate results as new events arrive.

After data enters AWS, it needs a storage service that matches how it will be used. Amazon S3 is a common foundation for machine learning data. It is object storage that can hold structured files, documents, images, audio, video, model artifacts, and many other objects. It scales to very large datasets and commonly supports data lakes and training workflows.

Amazon EBS and Amazon EFS serve different needs. EBS provides block storage that behaves like a disk attached to compute. EFS provides a managed file system that multiple resources can share.

Operational information may already live in databases. Amazon RDS supports relational data and SQL queries.

The correct storage choice depends on data structure, access frequency, required latency, performance, durability, compliance, and cost.

The way data is stored inside files also changes how efficiently it can be processed. CSV is a simple format for tabular data. It is easy for people to view and widely supported, but it contains limited information about data types. Large CSV datasets can use significant storage and require a system to read complete rows even when only a few columns are needed.

JSON is useful for events, APIs, and semi-structured information. It supports nested fields and flexible records.

Traditional business systems often focus on rows and columns, but modern AI systems use a wider range of data. Structured data includes transactions, account details, product records, sensor measurements, and other fields with a defined schema. This type of data often comes from relational databases, data warehouses, or operational applications.

A dependable ingestion pipeline can be understood through five checkpoints: source, transfer, validation, storage, and observation. At the source, the team confirms availability, ownership, permissions, and schema.

To conclude Part 1, data collection creates the foundation for every later machine learning decision. Batch ingestion supports scheduled or grouped processing, while streaming supports continuously arriving events and faster reactions.

Thank you so much for watching, see you in the next one.